Chest X-ray Generation Using VAE

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1 Hyperparameter Settings and Model Training

We implement a convolutional variational autoencoder (VAE) in PyTorch to generate synthetic chest X-ray images. The Chest X-Ray Images (Pneumonia) dataset contains NORMAL and PNEUMONIA images. All images are converted to grayscale, resized to 128×128, and normalized to $[0, 1]$. We use 4,695 images for training, 521 for validation, and 624 for testing.

The VAE encoder contains four convolutional layers with 32, 64, 128, and 256 channels. It maps each image into a 64-dimensional latent distribution. A symmetric decoder reconstructs images from sampled latent vectors.

The training objective combines reconstruction loss and KL divergence:

$$\mathcal{L}_{VAE} = \mathcal{L}_{rec} + \beta \mathcal{L}_{KL}. \tag{1}$$

We use mean squared error (MSE) for reconstruction and set $\beta = 1$. The model is trained for 25 epochs using Adam, with a batch size of 32 and learning rate of 0.001. The checkpoint with the lowest validation loss is selected for evaluation.

2 Training Loss Curves

Figure 1 shows the training and validation losses. The total training loss decreases from 396.49 to 110.60 over 25 epochs, indicating successful convergence. Reconstruction loss and KL divergence also decrease during training.



Figure 1: Total VAE loss, reconstruction loss, and KL divergence.

The validation loss follows a similar decreasing trend, with occasional fluctuations. The best validation loss of 110.27 occurs at epoch 24, which is selected as the final model checkpoint. No clear overfitting trend is observed.

3 Inception Score and FID Evaluation

We evaluate image generation using Fréchet Inception Distance (FID) and Inception Score (IS). A total of 624 images are generated from random latent vectors and compared with 624 real test images.

Both metrics use an ImageNet-pretrained InceptionV3 network. FID uses 2,048-dimensional pooled features, while IS uses predicted class probabilities.

The evaluation results are:

FID: 262.4125 Inception Score: 1.9429

The FID indicates a noticeable difference between real and generated image distributions, while IS suggests limited diversity under the pretrained classifier. Since InceptionV3 is trained on natural images, these metrics may not fully reflect medical image quality.

4 Generated Image Examples

Figure 2 shows 16 chest X-ray images generated by sampling latent vectors from $z \sim \mathcal{N}(0, I)$.

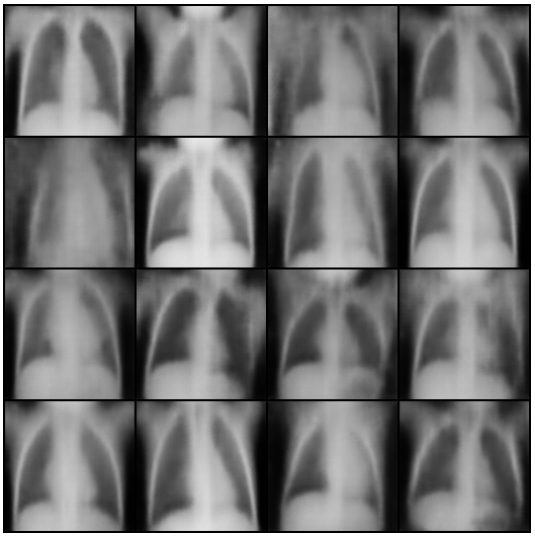


Figure 2: Chest X-ray images generated by the trained VAE.

The generated images contain recognizable lung shapes and central chest structures. However, they appear blurry and lack fine anatomical details, such as ribs and lung textures. Some images also exhibit similar patterns, indicating limited diversity.

Overall, the VAE successfully learns general chest X-ray structures and generates new images, but improvements are needed to produce sharper and more realistic samples.

Code and Results: GitHub Repository: witnesszc23/ECE-285